Every process function lifts to a unitary process

Draft note. Not submitted. Not refereed. Canonical source of the paper; `lift.pdf` is built from this file.

Abstract

A process function is a classical map w from the outputs of several parties to their inputs such that every choice of local deterministic interventions has exactly one fixed point. Baumeler, Costa, Ralph, Wolf and Zych showed that every process function extends to a bijection by adding a source and a sink, and left open whether every such bijection quantizes to a valid unitary quantum process; Daher Ahmed and Kunjwal (2026) state this as a conjecture and prove it for a subclass. We give a short argument that it holds for every process function with finite alphabets: the permutation unitary $U|o,e\rangle = |w(o)+e, o\rangle$ composed with arbitrary local unitaries induces a unitary from source to sink. The key combinatorial fact, that two distinct output strings always disagree at some party whose input w assigns equally, is the pairwise-exclusivity property of Dourdent et al. (2026). The philosophical sections read the result as saying that fixed-point consistency is already a quantization condition. The result is a draft: the exclusivity lemma and the identity $M M^\dagger = 1$ (complex amplitudes, any finite number of parties, any finite alphabets, local ancillas) are machine-checked in Lean 4 with Mathlib (theorems PFUL.induced_mul_conjTranspose, PFUL.induced_conjTranspose_mul and PFUL.induced_unitary, the last stating membership in the unitary group; standard axioms only). The step from "unitary under local unitaries" to "valid process" (Appendix A) is also machine-checked: for arbitrary local CPTP maps in Kraus form via their Stinespring isometries (PFUL.valid_kraus), for instruments (PFUL.valid_instrument), and in the Choi/process-matrix form $\mathrm{Tr}[W(\rho \otimes 1 \otimes C_A \otimes ...)] = 1$ for all local CPTP Choi matrices, with party ancillas (PFUL.Choi.bornC_anc_eq_one). The two forms are proved equivalent (PFUL.Choi.afnb_iff_density), and Choi's theorem is proved too (PFUL.Choi.isCP_iff_choi); PFUL.Choi.validity_package bundles the chain. Appendix F extends the theorem to countable alphabets, in the paper and in Lean (PFUL.Countable.induced_unitary_l2): exclusivity gives orthonormal rows, and unit columns follow because the column norm is the expected number of fixed points of a random local intervention, which is 1. The Choi form in infinite dimensions is not formalised, and continuous alphabets remain open. A second part studies stochastic classical processes. Mixtures of process functions are purifiable, and we conjecture that nothing else is. We give exact non-purifiability certificates for E_{ex1} and two vertex orbits, several structural lemmas, a counterexample to a natural strengthening, a no-go for certificates that use only the fixed source state, the complete facet list of the 3-party binary process-function hull (29,344 facets in 93 symmetry orbits), a negative result for symmetry-reduced moment relaxations (Appendix E: feasible for E_{ex1}, hence weaker than the inertia certificate), and numerical evidence. The conjecture itself remains open; Section 9 states exactly what is proved.

1. Result and short proof

Setting. Parties $k = 1..n$. Party k has a finite input alphabet I_k, identified with the cyclic group $Z_{|I_k|}$, and a finite output alphabet O_k. A process function is $w = (w_k)$: $\mathrm{prod}\, O_k \to \mathrm{prod}\, I_k$ such that for every tuple of local functions $f_k : I_k \to O_k$ the map $o \to (f_k(w_k(o)))_k$ has exactly one fixed point. The source has output space $P = \mathrm{prod}\, I_k$, the sink has input space $F = \mathrm{prod}\, O_k$, and

$$U\,|o\rangle_O\,|e\rangle_P = |w(o)+e\rangle_I\,|o\rangle_F \qquad \text{(componentwise addition mod } |I_k|).$$

U is a permutation of basis states, hence unitary, for every map w (Baumeler et al. 2019, Theorem 2).

Lemma (exclusivity). If w is a process function and $s \neq s'$ in $\prod O_k$, there is a party k with $s_k \neq s'_k$ and $w_k(s) = w_k(s')$.

Proof. Otherwise, for every k, $w_k(s) = w_k(s')$ implies $s_k = s'_k$. Define $f_k(x) = s_k$ if $x = w_k(s)$, $f_k(x) = s'_k$ if $x = w_k(s')$ (and arbitrary otherwise); the two clauses agree when $w_k(s) = w_k(s')$. Then s and s' are two distinct fixed points of the same intervention, a contradiction. (This is the forward direction of Dourdent et al. 2026, Theorem 7.)

Theorem. Let V_k be any unitary from $I_k \otimes X_k$ to $O_k \otimes Y_k$, with local ancilla spaces X_k, Y_k ($|I_k||X_k| = |O_k||Y_k|$). Composing U with the V_k (feeding each I_k into V_k and each O_k back into U) gives the operator M from $P \otimes X$ to $F \otimes Y$ with entries

$$M[(s,y),(e,x)] = \prod_k V_k[(s_k, y_k), (w_k(s) + e_k, x_k)].$$

M is unitary.

Proof. $(M\,M^\dagger)[(s,y),(s',y')]$ = sum over e, x of the product over k, which factorises as $\prod_k \sum_{e_k, x_k} V_k[(s_k,y_k), (w_k(s)+e_k, x_k)]\ \mathrm{conj}\ V_k[(s'_k,y'_k),(w_k(s')+e_k, x_k)]$. If $w_k(s) = w_k(s')$, substituting $i = w_k(s) + e_k$ turns the k-th factor into the inner product of rows (s_k,y_k) and (s'_k,y'_k) of the unitary V_k, i.e. a Kronecker delta. If $s = s'$ every factor is of this kind, giving $\mathrm{delta}_{y,y'}$. If $s \neq s'$, the Lemma gives a party with $w_k(s) = w_k(s')$ and $s_k \neq s'_k$, whose factor is 0. So $M\,M^\dagger = 1$, and M is square, hence unitary.

In the language of Araújo, Feix, Navascués and Brukner (2017, Definition 1), the process $|U\rangle\rangle\langle\langle U|$ is therefore *pure*. Section 2 spells out why this implies validity for arbitrary local CPTP maps and instruments, including ancillas shared or entangled with outside systems. Fixing the source to $|0\rangle$ and tracing the sink recovers the diagonal process matrix of w.

Converse (numerical only). If w is not a process function, the exclusivity property fails (Dourdent et al. 2026, Theorem 7); we have not proved that unitarity then fails in general, but numerically (code/check.py) every non-process function in the 2- and 3-party binary cases fails unitarity for random local unitaries. The "open/closed" contrast is the point: U is a bijection for every w; closing the loop is unitary exactly when w is consistent.

Checks. code/check.py enumerates all 12 binary 2-party and 744 binary 3-party process functions, verifies $\|MM^\dagger - 1\| < 1\text{e-}13$ for Haar-random local unitaries (with and without qubit ancillas) and for ternary 2-party examples, and verifies that non-process functions fail. Lean status is in lean/STATUS.md.

2. From unitarity to validity

Section 1 shows that the lift is *pure* in the sense of Araújo et al. (unitaries in, unitary out). Appendix A derives full process validity from this (any local CPTP maps, local and shared/entangled ancillas, instruments) using Stinespring dilation. Appendix B covers input and output alphabets of different sizes. Appendix C treats stochastic

classical processes: mixtures of process functions are purifiable, but a non-deterministic extremal point of the Baumeler–Wolf polytope provably is not (exact certificate).

3. Backward causation, before the formalism

The linguistic objection to backward causation is that a cause is, by definition, earlier than its effect. Dummett argued in 1954 and again in 1964 that this is a stipulation about the word, not a discovery about the world. Black's bilking argument is the better objection. If an earlier event is supposed to be the effect of a later one, an agent who sees the earlier event can intervene so as to prevent the later one. Either the agent cannot intervene, which wants an explanation, or the later event was not necessary for the earlier one. Lewis's reply splits the modal: one can, relative to the local facts and the laws, and one cannot, relative to the whole past. That dissolves a verbal paradox. It does not say which histories the laws admit.

General relativity makes the question physical. Gödel's 1949 solution has closed timelike curves. Morris, Thorne and Yurtsever showed that a traversable wormhole, if one could be held open, can be converted into a time machine by relative motion of the mouths. The Cauchy problem on the resulting spacetime is the grandfather paradox in differential equations. Friedman, Morris, Novikov, Thorne and collaborators proposed, in 1990, that only self-consistent solutions occur. Echeverria, Klinkhammer and Thorne then integrated the hard-sphere billiard on a wormhole and found initial data with one consistent continuation, data with two, and data for which none was apparent. Carlini, Frolov, Mensky, Novikov and Solodukhin derived a version of self-consistency from stationarity of the action in a model, and found the same underdetermination. Novikov's principle removes inconsistent solutions. It does not choose among consistent ones. Earman's survey of the classical problem remains the right warning: a consistency condition that is imposed after the dynamics is not a dynamics.

4. Three quantum replies, and why none is the criterion

Deutsch's 1991 model asks for a fixed point of the partial trace around the curve. In finite dimension a fixed point exists, so no particular action is forbidden, and the grandfather paradox becomes a mixed state. The cost is nonlinearity. The model clones unknown states. Aaronson and Watrous showed that the computational power is PSPACE. Bennett, Leung, Smith and Smolin argued that the power is an artefact of demanding a fixed point for a distribution that an ordinary preparation would not produce. Tolksdorf and Verch showed that the Deutsch condition can be met to arbitrary precision in quantum field theory on a spacetime with no closed timelike curve at all. A Deutsch fixed point is not diagnostic of a loop.

Lloyd, Maccone and collaborators replace the fixed point by post-selection, the P-CTC model. Inconsistent branches are the ones that fail the post-selection. The grandfather paradox is resolved by the experiment not happening. Process functions sit inside this class: the event list appears conditional on the post-selection succeeding. That is a simulation. It is not an account of why the inconsistent branch was never a physical option. Post-selection is a filter on samples, of the same shape as a measure-zero axiom.

The process-matrix formalism of Oreshkov, Costa and Brukner is the third reply, and the one this note uses. A process is a map from local instruments to probabilities, required to give a probability for every choice of instruments, with no global causal order assumed. Classical deterministic processes in that class are process functions. Baumeler and Wolf described the polytope. Three parties suffice for a violation of a causal inequality. The Lugano function, found by Araújo and Feix and studied by Baumeler and Wolf,

```
w(a,b,c) = (not b and c, not c and a, not a and b),
```

is the standard witness. Each party's input depends on the other two outputs. The dependence graph is a cycle. Every deterministic intervention has exactly one fixed point. Two parties are not enough: the twelve binary two-party process functions are all causally ordered. Tobar and Costa extended the characterisation to any number of parties.

5. What the fixed-point clause is doing

The clause is an attempt at the criterion the filters do not give. A map from outputs to inputs is admitted only when every local function, freely chosen, composes with it to give exactly one fixed point. No fixed point is the grandfather case: the intervention has nowhere to land. More than one is the bootstrap: the loop underdetermines its own contents. The freedom is the content of the no-new-physics principle in Baumeler, Costa, Ralph, Wolf and Zych. Any operation possible in an ordinary region remains possible in a region that does not itself contain the closed curve. Consistency is not allowed to forbid the operation. It is allowed only to constrain the global solution.

Dourdent, Leitherer, Boghiu, Simonov, Kunjwal and Acín have since said the same thing without quantifying over interventions. A list of events is the event list of a process function exactly when it is complete and pairwise exclusive. If any two of several questions can be answered jointly, so can all of them. That is a multipartite form of Specker's principle. It is a constraint on the process, not an extra axiom that paradoxical histories have measure zero. Cyclic causation is not the enemy. Cyclic signalling is.

6. Reversibility, and the quantization question

Reversibility is not free on the original alphabet. The Lugano function is two-to-one. Of the 256 binary two-party maps, twelve are process functions and none equals its input-output reverse. Baumeler, Costa, Ralph, Wolf and Zych, Theorem 2, restore reversibility by a source, a region with trivial input, and a sink, a region with trivial output. The extended process can be read backwards. The form used here is the permutation unitary

```
U |o, e⟩ = |w(o) + e, o⟩.
```

The inverse reads the sink as the output string and the source as the difference between the input and w(o). A bijection of a finite set is a unitary. A unitary on a fixed basis is not yet a quantum process. Parties must be able to insert instruments that are not diagonal in that basis. A Hadamard is the test case. If the induced map from source to sink ceases to be trace-preserving, the classical consistency clause has not lifted, and quantum theory needs an extra filter. If it remains trace-preserving for every instrument, the fixed-point clause was already the quantum consistency clause.

Daher Ahmed and Kunjwal conjectured, on 30 September 2026, that every process function purifies to a unitary process. They prove it under mutual exclusivity of control conditions, and under a second sufficient condition, unambiguity, and they note that the bare permutation unitary of the reversible extension may fail to be a valid process. The argument below is that it does not fail. Araújo, Feix, Navascués and Brukner had already written down, in 2017, a purification of the Lugano process by exactly this construction (the standard $|x,y\rangle \to |x, y + f(x)\rangle$ trick, credited there to Baumeler and Wolf); a routed-circuit reconstruction is due to Vanrietvelde, Ormrod, Kristjánsson and Barrett. Baumeler, Gilani and Rashid (2022) proved that the diagonal (dephased) process matrix of every process function is valid, and remarked, without a separate proof for the coherent operator, that the

reversible extensions are therefore unitarily extensible. The contribution claimed here is a short general proof for the coherent permutation unitary, for every process function.

7. Time symmetry and retrocausality

Price has argued that a time-symmetric ontology for quantum theory should be retrocausal, on the assumption that the quantum state is real. Leifer and Pusey replaced that assumption with lambda-mediation: correlations between a preparation and a later measurement are mediated by the ontic state of the system. No retrocausality, operational time symmetry, and lambda-mediation then imply a timelike Bell factorization, which sequential measurements violate. Maudlin's reply is that the operational time-reverse of an experiment is not always itself an experiment of the same kind. The no-go is conditional.

The dilation in this note is adjacent to that argument, not a solution of it. A process function that violates a causal inequality already has each party's input depending on another party's output. The later setting is an argument of the earlier input. Lambda-mediation, in the strict past-screening sense, fails because of the cycle, not because a retrocausal hidden variable was added by hand. Read forwards, the source is an input and the sink is an output. Read backwards, they exchange roles, and the inverse of the permutation unitary is the reversed process. The objection that the bare Lugano function is not an involution on its own three bits is an objection to refusing the source and the sink. Whether the dependence is the retrocausality Price asked for, or only an operational time-reverse, is a reading of the same operator. The mathematics does not choose the reading. It does remove one cheap reply: one cannot say that the cyclic classical model dies when instruments become unitary.

8. What this does not settle

Embeddability is untouched. A unitary process on a finite alphabet is not a solution of Einstein's equation, and it is not a local Hamiltonian flow on a wormhole. The billiard has the wrong multiplicity. Post-selection realises the event list conditional on the post-selection succeeding. The theorem says the list has a unitary dilation. It does not say the dilation is a spacetime.

Purifiability in the sense of open quantum problem 43, whether every extensibly causal process is purifiable, is related but not identical. The theorem puts every process function, including those that violate causal inequalities, inside the unitarily extendible fragment. It does not classify processes that are not deterministic classical functions.

Freedom is not settled. Nothing here decides whether an agent whose output is an argument of an earlier input has a choice. The fixed-point clause says the choice, if made, completes uniquely. It does not say the choice was open. Whether this is an adequate reply to Black depends on whether consistency conditions may rule out interventions, or only rule out histories given the interventions. The note takes the second reading: consistency rules out histories given the interventions, never the interventions themselves (Section 5), and the formalism is built for it.

The combinatorial lemma is finite for any fixed alphabet and is the piece an attacker has to break. The linear algebra after it is row orthogonality. See lean/STATUS.md for exactly what is and is not machine-checked.

9. Status of the purification conjecture

Conjecture: a classical process (a point of the Baumeler-Wolf polytope) is purifiable in the sense of Araujo et al. (2017) if and only if it is a mixture of process functions.

Proved.

- Every process function lifts to a unitary process (Section 1). The core identity is machine-checked in Lean (https://github.com/drewarrowood/process-function-unitary-lift/blob/main/lean/STATUS.md). Validity follows by Stinespring (Appendix A); the whole validity chain is machine-checked in finite dimensions (PFUL.valid, valid_kraus, valid_instrument, isCP_iff_choi, afnb_iff_density_split, bornAnc_eq_one, bundled in PFUL.Choi.validity_package).
- Countable alphabets (Appendix F): the lift is unitary on l^2, in the paper and in Lean (PFUL.Countable.induced_unitary_l2), via the expected-number-of-fixed-points argument for the column side. Not formalised: the Choi form in infinite dimensions. Open: continuous alphabets (F.5).
- Mixtures of process functions are purifiable: use a coherent random source. Convex combinations of purifiable processes are purifiable (Appendix C).
- Classical version (D.6): a classical process is the marginal of a classical reversible process with a random source if and only if it is a mixture of process functions. So the conjecture says that quantum purifiability gives nothing beyond classical purifiability, for classical processes.
- Exact non-purifiability certificates (rational arithmetic) for E_{ex1} and for vertex orbits 7 and 50 (Appendix C; pipeline in code/stochastic/cert).
- Structure of any purification: the sink records (o, i) orthogonally (D.1). Constraints on the fixed source state are vacuous (D.2). Validity under quantum-controlled interventions holds for every classical process (D.5).
- A natural strengthening, a single classical source basis (D.3), is false. Counterexample: a causal circuit (D.4).
- No-go (D.8): no certificate that uses only the fixed source sector can rule out purification of any classical process.

Evidence.

- None of the 49 remaining non-deterministic vertex orbits lifted under a GPU purification search. Best residuals were far from zero (Appendix C).
- The process-function hull for 3 binary parties has exactly 29,344 facets in 93 orbits. A single orbit (82) is violated by E_{ex1} and by all 51 sampled non-deterministic vertices (D.7).
- Along the segment from Lugano to E_{ex1}, purifiability appears to stop exactly where the hull does (D.8).

Open. The quantum case. Any proof must use the purifying unitary on source states orthogonal to the source state. That is the non-convex isotropy condition of AFNB Theorem 4, and it is where every attempted certificate (single-form inertia, block forms, SDP relaxations) has stalled for the larger orbits. The degree-2 moment relaxation over $P \otimes P$ is even weaker than single-form inertia: it is feasible for E_{ex1} (Appendix E).

Interpretation (philosophical, not a result). Consistent time loops are the setting: a process function is a loop in which every local choice has exactly one consistent history. The conjecture asks whether chance inside such a loop can be fundamental, or only ignorance about which deterministic loop obtains. If it holds, every reversible quantum loop with classical statistics is a coherent ignorance mixture of deterministic consistent loops. Genuinely chancy loops like E_{ex1} would then be classically consistent but not reversibly realisable. Reversibility, not consistency alone, would be the constraint that excludes basic chance from loops. D.4 adds a caveat: the "which-loop" variable can be contextual, so the ignorance reading is about W, not about any single hidden label.

Appendix A. Unitarity under local unitaries implies validity

Definitions (AFNB 2017, Sec. 2). A process with source P, sink F and parties k is an operator $W \geq 0$ on $P \otimes F \otimes \prod_k (I_k \otimes O_k)$. For local maps $A_k : L(I_k \otimes X_k) \to L(O_k \otimes Y_k)$, the induced map is the link product $G_A = W * (A_1 \otimes ... \otimes A_n)$, a map $L(P \otimes X) \to L(F \otimes Y)$ with $X = \otimes_k X_k$ and $Y = \otimes_k Y_k$. W is *valid* if G_A is CPTP whenever every A_k is CPTP, for all finite-dimensional local ancillas X_k, Y_k. W is *pure* (their Definition 1) if G_A is unitary whenever every A_k is unitary. Their Theorem 2 says W is pure iff $W = |U\rangle\rangle\langle\langle U|$ for a unitary U. For pure Choi vectors the link product composes operators: if $A_k(\text{rho}) = V_k \text{ rho } V_k\dagger$, then $G_A(\text{rho}) = M \text{ rho } M\dagger$, where M is the operator of Section 1.

Claim. $W = |U\rangle\rangle\langle\langle U|$ with U the permutation lift of a process function w is a valid process.

Step 1 (local unitaries). By the Theorem of Section 1, G_A is conjugation by a unitary M whenever every A_k is conjugation by a unitary V_k. This holds for any ancilla dimensions with $|I_k||X_k| = |O_k||Y_k|$. (Machine-checked, both sides, including the dimension count: `PFUL.induced_mul_conjTranspose`, `PFUL.induced_conjTranspose_mul`, `PFUL.induced_unitary`; see lean/STATUS.md.)

Step 2 (local CPTP maps, via Stinespring). Let $A_k : L(I_k\ X_k) \to L(O_k\ Y_k)$ be CPTP. Stinespring dilation gives an isometry $S_k : I_k\ X_k \to O_k\ Y_k\ E_k$ with $A_k(\text{rho}) = \text{Tr}_{E_k}[S_k \text{ rho } S_k\dagger]$. Pad the input with E'_k of dimension $|O_k||Y_k||E_k|$ and the output with E''_k of dimension $|I_k||X_k|$. The two sides then have equal dimension, and the isometry $\text{psi} \otimes |0\rangle \to (S_k \text{ psi}) \otimes |0\rangle$, defined on the $|0\rangle$-slice, extends to a unitary $V_k : I_k\ X_k\ E'_k \to O_k\ Y_k\ E_k\ E''_k$ by completing orthonormal bases. So $A_k(\text{rho}) = \text{Tr}_{E_k\ E''_k}[V_k\ (\text{rho} \otimes |0\rangle\langle 0|_{E'_k})\ V_k\dagger]$. The ancillas E', E, E'' never enter U, and the link product is multilinear and commutes with appending fixed states and with partial traces on systems it does not contract. Hence

$$G_A(\text{rho}) = \text{Tr}_{E\ E''}[\ M\ (\text{rho} \otimes |0\rangle\langle 0|_{E'})\ M\dagger\ ],$$

where M is the operator of Step 1 for the unitaries V_k, with enlarged local ancillas $X_k\ E'_k$ and $Y_k\ E_k\ E''_k$. Appending a state, a unitary conjugation and a partial trace are each CPTP, so G_A is CPTP.

Step 3 (shared and entangled ancillas). An ancilla R that is shared between parties, or entangled with any outside system, is covered by complete positivity. The joint state lives on $P\ X\ R$; $G_A \otimes \text{id}_R$ is CPTP because G_A is; outputs are normalised states, and every probability that a later measurement on $F\ Y\ R$ assigns is non-negative and sums to 1. Pre-shared entanglement between laboratories is the special case in which part of X_j and part of X_k start in an entangled state supplied through the source. It is also covered, because Step 2 already allows arbitrary input states on $P \otimes X$.

Step 4 (instruments). Let $\{A_k^{a_k}\}$ be CP maps summing to a CPTP map A_k. The link product of positive operators (Choi operators) is positive, so each $G_a = W * \otimes_k A_k^{a_k}$ is CP. Multilinearity gives $\sum_a G_a = G_A$, which is CPTP by Step 2. Then $p(a) = \text{Tr } G_a(\text{rho}) \geq 0$ and $\sum_a p(a) = 1$ for every input rho. This is the validity condition of Oreshkov, Costa and Brukner, applied to the process seen by the parties once the source is prepared and the sink discarded. Fixing the source in $|0\rangle$ and tracing the sink yields $W_w = \sum_o |o\rangle\langle o|_O \otimes |w(o)\rangle\langle w(o)|_I$, the classical process matrix of w.

So purity gives validity with no further assumption. The only input specific to process functions is Step 1.

Lean (finite dimensions). All four steps are machine-checked, standard axioms only.

- *Step 1, isometries:* `PFUL.induced_isometry`. If every local map is an isometry ($V_k^\dagger V_k = 1$, not necessarily square), then $M^\dagger M = 1$. It rests on the column identity with constant row sums (`PFUL.Column.sum_prod_eq_prod`).
- *Step 2, Stinespring:* `PFUL.krausDilP_isometry` and `PFUL.ptrace_krausDil`. Any Kraus family has the isometric dilation $V = \sum_j K_j \otimes |j\rangle$, which reproduces the channel. `PFUL.Choi.kraus_of_choi` gives a Kraus family from any PSD Choi matrix (spectral theorem), complete when the map is TP.
- *Step 3, ancillas:* `PFUL.valid_kraus` (with `PFUL.valid` for unitaries). For any joint density matrix of source and ancillas (entangled or shared), the global output and the sink state are density matrices.
- *Step 4, instruments:* `PFUL.valid_instrument`. Outcome-conditioned sink states are PSD and sum to a density matrix, so the probabilities sum to 1.
- *Process-matrix form:* `PFUL.Choi.procMat` defines $W = |U\rangle\rangle\langle\langle U|$. `PFUL.Choi.born_eq_trace` shows $\mathrm{Tr}[W (\rho \otimes 1_F \otimes \otimes_k C_k)^T] = \mathrm{Tr}[M \rho M^\dagger]$. `PFUL.Choi.bornC_eq_one` gives the AFNB/OCB normalisation = 1 for every source state and all local CPTP Choi matrices.
- *Ancillas, converse, Choi's theorem:* `PFUL.Choi.born_anc_eq_one` and `bornC_anc_eq_one` give the Choi-form normalisation with party ancillas (local maps $I_k X_k \to O_k Y_k$, ancilla state folded into the source state). `PFUL.Choi.procMat_posSemidef` shows W is PSD. `PFUL.Choi.afnb_iff_density` shows AFNB normalisation is equivalent to the density-matrix condition. `PFUL.Choi.isCP_iff_choi` is Choi's theorem (CP iff PSD Choi matrix), so CP is a theorem rather than a definition.

Bookkeeping: `PFUL.Choi.bornAnc_eq_one` needs no structure on the ancilla types. `PFUL.Choi.bornSplit_eq_one` and `PFUL.Choi.afnb_iff_density_split` use exactly the index layout of `ValidWith`. `PFUL.Choi.validity_package` bundles unitary validity, Kraus-channel validity and AFNB normalisation in one theorem. Not formalised: the Choi form in infinite dimensions, which needs trace-class operators and partial traces on l^2 that Mathlib does not yet have. The unitary layer there is `PFUL.Countable.induced_unitary_l2`.

Appendix B. Unequal input and output alphabets

Nothing in Section 1 needs $|I_k| = |O_k|$.

- $U : O \otimes P \to I \otimes F$, with $P = \prod I_k$ and $F = \prod O_k$, maps $|o,e\rangle$ to $|w(o)+e, o\rangle$. Both sides have dimension $\prod |I_k||O_k|$, and U is a permutation of basis states, so U is unitary for every w. Only I_k needs a group structure ($Z_{|I_k|}$); O_k is an arbitrary finite set.
- The exclusivity Lemma and its proof never compare I_k with O_k.
- A party with $|I_k| \neq |O_k|$ cannot apply a unitary $I_k \to O_k$. Its most general deterministic operation is a CPTP map, or a unitary with ancillas, $I_k X_k \to O_k Y_k$ with $|I_k||X_k| = |O_k||Y_k|$. The Theorem is stated in exactly that

generality (and so is the Lean theorem, whose local types I k, O k, X k, Y k are independent). The Gram computation gives M M† = 1, and M is square because dim(P X) = prod $|I_k||X_k|$ = prod $|O_k||Y_k|$ = dim(F Y). Appendix A then applies verbatim.

What holds. For any finite input and output alphabets, with any sizes, the permutation lift of a process function is a pure and valid process. **What is not covered here.** Infinite alphabets (countable ones are treated in Appendix F; continuous ones remain open, and Baumeler et al. 2019 also treat continuous variables), and lifts built from a different bijection than U.

Appendix C. Stochastic classical processes

Question. Does every logically consistent classical process admit a unitary extension, in the sense of AFNB purifiability? The process may be stochastic, not just a deterministic process function.

What is known (verified sources).

- Baumeler and Wolf (NJP 18, 013036 (2016), https://arxiv.org/abs/1507.01714) show the classical processes form a polytope. With two binary parties it has 12 extremal points, all deterministic. With three binary parties it has 710,760 extremal points, of which only 744 are deterministic, i.e. process functions. The rest are "proper mixtures" of logically inconsistent deterministic processes. One explicit example is E_{ex1} = (C + C̄)/2, the uniform mixture of the circular identity channel C ($x_1 = a_3$, $x_2 = a_1$, $x_3 = a_2$) and its bitwise negation C̄.
- Araújo, Feix, Navascués and Brukner (Quantum 1, 10 (2017), https://arxiv.org/abs/1611.08535) define purifiability (their Definition 3). Their Theorem 4 is an exact criterion, and Section 5 gives a necessary condition, dim $V_W \geq$ rank W.
- Baumeler, Gilani and Rashid (Quantum 6, 673 (2022), https://arxiv.org/abs/2104.06234) assert unitary extensibility for process functions only, via reversible embeddings, without a separate proof.
- Daher Ahmed and Kunjwal (https://arxiv.org/abs/2610.00579) conjecture that process functions are purifiable. Tobar and Costa (CQG 37, 205011 (2020), https://arxiv.org/abs/2001.02511) treat reversible deterministic dynamics.
- I found no source that claims to settle purifiability of the non-deterministic extremal points.

C.1 Proposition (proved). Every convex mixture of process functions is purifiable.

Proof. Let W = $\sum_j p_j W_{w_j}$. Take source P' = $C^J \otimes$ P and sink F' = $C^J \otimes$ F, and define U|o, e, j⟩ = $|w_j(o) + e\rangle_I |o, j\rangle_{F'}$. With local unitaries inserted, the induced operator is the block sum $\oplus_j M_j$. Each M_j is unitary by the Theorem of Section 1, so the process is pure. Feed $\sum_j \sqrt{p_j}|j\rangle \otimes |0\rangle$ into P'. This is a fixed pure state, equivalently $|0\rangle$ followed by a unitary that is absorbed into U. Then trace out F'. Because the sink keeps a copy of j, the cross terms vanish and the result is $\sum_j p_j W_{w_j}$ = W. ∎

The control register must enter through the source and leave through the sink. If it is not handed to the sink, the coherence between branches survives, and the result is no longer the classical mixture. Hence the whole deterministic-extrema polytope of Baumeler and Wolf is purifiable. This covers every two-party classical process with binary inputs and outputs, since there all extremal points are deterministic.

C.2 E_{ex1} is not purifiable (exact certificate). Scripts are in code/stochastic/.

Reduction. In AFNB Theorem 4 the auxiliary sink F' behaves as a party with trivial output. Every term of the projector $L_V^{\perp}$ therefore traces out and replaces F'. Write $w_i = \Sigma_k |w_{i,k}\rangle|k\rangle_{F'}$ and define X_i by $X_i |p_k\rangle = \mathrm{sqrt}(E_{p_k}) |w_{i,k}\rangle$. The conditions become $L_3^{\perp}(X_i \Lambda X_j\dagger) = 0$ for all i, j, with $\Lambda = \mathrm{diag}(1/E)$ on the support. Here $L_3^{\perp}$ is the ordinary three-party projector on 64×64 matrices, and $X_0 = W$. This takes seconds instead of tens of minutes, and it reproduces the full computation (dim V_W 27 for E_{ex1} and 56 for Lugano).

Certificate.

- E_{ex1} is logically consistent and is a vertex of the three-party binary polytope (active-constraint rank 64 of 64). Hence it is not a mixture of process functions.
- rank W = 16. Exact integer arithmetic (python-flint) gives dim $V_W = 27$.
- One explicit integer combination of the constraint functionals (code/stochastic/exact.py, cert_phi.npy) gives a sesquilinear form on V_W. Its symmetric part has exact inertia (13 positive, 13 negative, 1 zero), computed by rational congruence elimination.
- The span of w_0, ..., w_{15} is a 16-dimensional subspace on which this form must vanish, so it is totally isotropic for the symmetric part. But a totally isotropic subspace has dimension at most 1 + min(13, 13) = 14 < 16. So no purification exists.
- An independent numerical check agrees. A gradient search for purifications reaches residual $8 \cdot 10^{-14}$ for Lugano (which is purifiable), but stalls at 79.7 for E_{ex1}.

C.3 Purifiable set and the conjecture. The control-register argument of C.1 works for any purifiable processes, not just process functions. So the purifiable classical processes form a convex set containing the hull of the process functions. We conjecture they are exactly that hull:

Conjecture. A classical process is purifiable iff it is a mixture of process functions.

Evidence so far.

- 40,000 random LP vertices of the three-party binary polytope gave 39,354 deterministic vertices and 646 non-deterministic ones. These fall into 51 orbits under party permutations and local input/output relabellings (code/stochastic/reps.json). Ranks are 15–23.
- AFNB's necessary condition dim $V_W \geq$ rank W holds for all 51 orbits (nec_partial.json), so on its own it decides nothing. dim V_W ranges from 27 to 256.
- Exact certificates of non-purifiability exist for the 2 orbits with dim $V_W = 27$, one of which contains E_{ex1} (exact2.py, cert_phi_7.npy, cert_phi_50.npy).
- For the other 49 orbits (dim $V_W \geq 82$) the random-form isotropy test was inconclusive. The GPU gradient search of C.4 found no purification for any of them (best residual ≥ 12.97). This is numerical evidence only.
- No purifiable non-mixture has been found. That is weak evidence, because the large cases are untested.
- All 1,488 mixtures $(q + E_{ex1})/2$ and $(q + 4\,E_{ex1})/5$, with q one of the 744 process functions, lie outside the process-function hull (LP check). They are untested candidates for non-extreme counterexamples.

C.4 GPU sweep over the 49 open orbits (numerical evidence, not proof).

- Method: code/stochastic/gpu/gpusearch.py ran on one RTX 3090. For each orbit it made 64 random restarts of batched L-BFGS in float32 on the reduced Theorem 4 residual, then polished in float64.

- Validation: the same code finds Lugano's purification (residual 1.2e-16) and stalls at 79.676 on E_{ex1}, which is certified non-purifiable.
- Result: no orbit reached the 1e-10 threshold. The best residuals range from 12.97 (orbit 40) to 59.35. Every orbit stalled far from zero.
- Interpretation: this is consistent with the conjecture that none of the 49 non-deterministic vertex orbits is purifiable. But a local search can miss solutions, so it is not a proof. Only orbits 7 and 50 (C.2) have exact certificates.

orbit	rank	dim V_W	best residual (float64)	restarts
0	15	124	20.12	64
1	16	176	13.03	64
2	15	156	15.62	64
3	15	156	15.49	64
4	16	128	21.93	64
5	16	160	12.98	64
6	16	144	18.48	64
8	16	82	35.84	64
9	16	112	27.06	64
10	16	82	34.75	64
11	16	128	21.72	64
12	16	112	26.91	64
13	16	196	12.97	64
14	16	144	18.49	64
15	16	168	13.02	64
16	15	156	19.57	64
17	15	164	15.82	64
18	16	160	13.02	64
19	16	160	12.97	64
20	16	82	35.32	64
21	16	128	21.73	64
22	16	164	18.44	64
23	16	172	12.99	64
24	16	112	26.78	64
25	16	136	21.76	64
26	16	128	21.88	64
27	16	144	18.5	64
28	15	124	20.12	64
29	16	144	18.51	64
30	16	168	13.02	64
31	16	160	13.02	64
32	21	160	49.61	64
33	16	164	18.46	64
34	20	174	35.72	64
35	16	160	13.02	64
36	16	160	12.97	64
37	16	160	12.98	64
38	22	176	48.56	64

orbit	rank	dim V_W	best residual (float64)	restarts
39	21	146	59.35	64
40	16	196	12.97	64
41	16	204	13.01	64
42	16	172	12.99	64
43	22	224	25.58	64
44	21	160	50.3	64
45	21	174	41.12	64
46	20	166	34.6	64
47	23	248	42.74	64
48	21	172	37.9	64
49	23	256	29.02	64

Proof idea for "only if" (not completed). A purification U restricted to classical permutation instruments gives a unitary M for every reversible local classical operation. One would like to show that M's action on basis states then decomposes into permutations, giving a deterministic decomposition. We have not proved this. E_{ex1} shows the classical consistency of W alone is not enough.

Answer. Not every classical process admits a unitary extension: E_{ex1} (and one other vertex orbit) provably does not. Mixtures of process functions do. Whether those are the only ones is the open conjecture above.

Appendix D. Purifications of classical processes: the physics route

Status labels: **[proved]**, **[conjecture]**, **[gap]**. Nothing here proves the conjecture; it is still open.

Setting. A purification of a classical process W (diagonal; equivalently a stochastic map p(i|o) on the joint classical inputs i and outputs o of the parties) is a unitary $V : P \otimes O \to F \otimes I$ together with a fixed source state $|psi\rangle$ in P such that (i) $W = Tr_F$ of the Choi vector of $V(|psi\rangle \otimes .)$, and (ii) for every choice of local operations (unitaries with ancillas) the induced map P ⊗ ancillas → F ⊗ ancillas is unitary (AFNB 2017). Write $V|psi, o\rangle = \sum_i |i\rangle \otimes |v_{o,i}\rangle$ with $v_{o,i}$ in F.

Lemma D.1 [proved]. $\langle v_{o',i'} \mid v_{o,i}\rangle = delta_{o\ o'}\ delta_{i\ i'}\ p(i|o)$. Proof. The Choi matrix of the induced channel O → I is $\sum_{o,o'} |o\rangle\langle o'| \otimes Tr_F[\ V|psi,o\rangle\langle psi,o'|V\dagger\ ]$, whose ((o,i),(o',i')) entry is $\langle v_{o',i'}|v_{o,i}\rangle$. W is diagonal with diagonal entries p(i|o). QED. Consequence: the sink holds a perfect record of the pair (o,i): the normalised vectors $f_{o,i} = v_{o,i}/\sqrt{p(i|o)}$ (over the support of W) are orthonormal, so dim F ≥ |supp W| = rank W = r (the same r as in AFNB Thm 4). Checked numerically on the source-sink lift of Lugano (code/stochastic/physics/lemma_d1_check.py).

Lemma D.2 [proved; negative]. Conditions (i)-(ii) restricted to the source state $|psi\rangle$ impose nothing beyond W being a classical process. For classical deterministic interventions with input copies, o = f(i) (+) a with the input i copied to a private ancilla, the output for ancilla value a is $\sum_i |i\rangle_{anc} \otimes \sqrt{p(i|f(i)+a)}\ f_{f(i)+a,\ i}$; by D.1 these are orthogonal for a ≠ a' and have total norm $\sum_i p(i|f(i)+a) = 1$ (logical consistency). Hence any non-purifiability (e.g. E_{ex1}, orbits 7 and 50) must come from the action of V on source states orthogonal to $|psi\rangle$. Any proof of the conjecture must use those states. This is where proposal (2) breaks down: the "which-function" label lambda cannot come from a basis of the span of $|psi\rangle$ alone.

Where proposal (2) fails [gap]. The intended argument: under each intervention the global map is a generalised permutation conditioned on a source label lambda; expanding $|\psi\rangle = \sum_{\lambda} c_{\lambda} |\lambda\rangle$ gives W = sum $|c_{\lambda}|^2 W_{w_{\lambda}}$. Three gaps: (a) Unitarity under local unitaries does not obviously make V, restricted to classical party inputs, a generalised permutation. D.1 gives orthogonality only for the fixed $|\psi\rangle$, and only after tracing the party wires through W. (b) Even when it holds for each intervention f separately, the label lambda may depend on f (no single basis of P works for all f). Ruling this out is exactly the noncontextuality-type step that is missing. (c) Coherences in $|\psi\rangle$ across labels: they vanish from W by D.1, but they constrain V on psi-perp, see (b).

Conjecture D.3. If V purifies a classical process, there is an orthonormal basis {lambda} of P such that $V|\lambda, o\rangle$ = phase * $|w_{\lambda}(o)\rangle \otimes |g_{\lambda}(o)\rangle$ for functions w_{λ}, with g_{λ} injective on outputs. D.3 implies the conjecture: each w_{λ} must then be a process function, since V must stay unitary under classical interventions restricted to the source state $|\lambda\rangle$, and by Baumeler-Wolf that means a unique fixed point. D.3 holds for our source-sink lifts. It has not been tested on the non-permutation purifications found by the optimiser (Lugano, App. C); that test is the next step. A counterexample to D.3 that is still a mixture would refute only D.3, not the conjecture.

D.4 Conjecture D.3 is false (update, Oct 10 2026) [proved, exact counterexample]

Write $G_{i,o} = V_o^\dagger (\Pi_i \otimes 1_F) V_o$ on P, where $V_o|s\rangle = V|s,o\rangle$. D.3 is equivalent to: all $G_{i,o}$ are projectors that commute pairwise (their common eigenbasis is the label basis lambda). Counterexample: a causally ordered circuit (code/stochastic/physics/d3_counterexample.py). Party A has trivial input and output oA. The source is P = m ⊗ a. Apply H^{oA} to m, CNOT m → a, send m to B's input, and route (oA, oB, a) to the sink. V is a permutation-and-Hadamard circuit of unitaries in a fixed causal order, so it is a valid unitary process. With psi = $|00\rangle$ the process is classical: $p(iB|oA=0) = \delta_{iB,0}$ and p(iB|oA=1) = 1/2. Lemma D.1 holds: W is diagonal. W is a mixture of process functions ($w_{\lambda}(oA) = \lambda * oA$). But the G's are Z-projectors for oA=0 and X-projectors for oA=1. They do not commute (commutator norm 1/2), so no source basis works. Restricting to psi does not help either: psi = $|00\rangle$ is an eigenvector of the oA=0 projectors only.

Consequence. Gap (b) is real. Even for causal processes, the "which-function" randomness is generated contextually, by measuring the quantum source in a basis that depends on earlier outputs. So no argument that extracts a single classical label from the source can work, and unitarity under all local unitaries does not force one. The mixture conclusion still holds here for a different reason: every causally ordered classical process is a mixture of causal deterministic ones. A proof of "purifiable ⇒ mixture of process functions" therefore has to show directly that W lies in conv(PF), for example that purifiability implies every facet inequality of the PF polytope. It cannot go through a source-label decomposition. Status: the conjecture is open; D.3 is refuted; no counterexample to the conjecture itself.

D.5 The hull route (Oct 10 2026)

Proposition D.5 [proved]: validity under quantum-controlled interventions is automatic, so it cannot be the criterion. Let W be any classical process, including E_{ex1} and all 49 open orbits. Let the parties apply arbitrary local instruments, coherently controlled by a shared ancilla that may be entangled across parties. Then the probabilities are normalised. Reason: W is diagonal, so it dephases every party's input and output. The effective intervention is then a classical joint behaviour q(o|i) produced from a shared quantum state by local operations, so

q is non-signalling. Every non-signalling behaviour is an affine (not necessarily convex) combination of local deterministic ones. The normalisation $\sum_{i,o} p(i|o)\, q(o|i)$ is affine in q and equals 1 on every local deterministic q (logical consistency), so it equals 1 for all such q. Positivity is termwise. So the proposed test "valid under quantum-controlled interventions" holds for every classical process. It cannot separate E_{ex1} from PF mixtures, and step (3) of the plan is moot. Purifiability is strictly stronger only because of its unitarity (reversibility) requirement.

Proposition D.6 [proved, sketch]: classical-reversible purifiability ⇔ mixture of PFs. Call W classically purifiable if it is the marginal, for a random source value lambda, of a classical reversible process: a bijection (lambda, o) → (i, sink) that stays a bijection P → F under every deterministic local intervention. For a fixed lambda this is a deterministic process. Bijectivity of the composite map under every intervention f forces exactly one consistent execution: zero leaves the source value with no image, and two collide by counting. So each w_{lambda} is a process function (Baumeler-Wolf 2016), and $W = \sum_{lambda} p(lambda)\, W_{w_{lambda}}$. The converse is the source-sink construction with a random source. Hence **the conjecture is equivalent to: a classical process that is quantum-purifiable is already classically purifiable.** D.4 shows the quantum purification itself need not be classical (its labels are contextual), so this has to be argued at the level of W.

Facets of the PF hull [computation]. The 744 process functions (3 parties, binary) span a 37-dimensional affine space. Facet enumeration results are recorded below. Each facet that some open orbit violates is a candidate inequality that purifiability would have to imply. Full facet enumeration was not done: pycddlib does not build on the box, and 744 vertices in dimension 37 may have very many facets. Instead, sep.py finds by LP a separating inequality of the PF hull (box-normalised, $|f| \le 1$): E_{ex1} and orbit 7 violate one by 6.0, orbit 8 by 4.0, and a PF mixture by 0, as expected. These inequalities are not reduced modulo the affine hull or by symmetry, so their supports carry no meaning yet. **No inequality has been derived from purifiability. The conjecture remains open.** Next steps: enumerate facets with lrs/normaliz or by symmetry-adapted LP, then try to derive one facet class from Theorem 4 (isotropy) together with Lemma D.1.

D.7 Facets of the PF hull: partial (Oct 10 2026)

Tools. lrslib 7.3 builds from source with gcc (no make, no GMP on the box; hybrid 64/128-bit plus built-in MP arithmetic). normaliz and PANDA were not installed: there is no make/cmake, conda or sudo on the box. The full enumeration with lrs (744 vertices in the 37-dim span, `facets/pf.ext`) produced no output in 5 minutes, not even from `estimates`, because the input is highly degenerate. **No facet count or runtime estimate yet.** A double description or symmetry-exploiting code (normaliz, PANDA with the 384-element symmetry group) on a rented multi-core CPU is the natural next step; its cost is unknown until a short benchmark is run.

Facets found by polar LP (`facets/facetlp.py`). Each LP optimum was checked to be a facet: its tight PFs have affine rank 36. Facets are classified up to the 384 symmetries (party permutations and input/output relabelings; all 384 preserve the PF set). Over E_{ex1} and the 51 sampled non-deterministic vertex orbits, the optimal separating facets fall into **8 classes**. Class 0 (418 tight PFs) is violated by E_{ex1} and 24 orbits, including orbit 8. Class 2 (418) covers orbits 7, 50 and 15 others; classes 1, 3-7 cover the rest. The class-0 facet, written with joint inputs i and outputs o (bits of parties A, B, C):

```
p(000|000) ≤ p(000|001) + p(010|001) + p(000|100) + p(010|110) +
p(011|111)
```

E_{ex1} gives 1/2 > 0. Combinatorial content for a process function w: if w(000) = 000, then w(001) is 000 or 010, or w(100) = 000, or w(110) = 010, or w(111) = 011.

Derivation from purifiability: not achieved. A natural attempt: use Lemma D.1 to read p(i|o) as squared norms of orthogonal sink vectors, then try to show the class-0 inequality follows from unitarity under suitable superpositions of the five interventions involved. This has not been carried through and is not claimed.

D.7 update: complete facet list (normaliz, Oct 10 2026)

Run: normaliz 3.8.9 (`-c -x=32`) on a rented 32-core vast.ai CPU instance, about 1.9 h, about $0.27 in total; the instance was then destroyed. Input `facets/pf.in` (744 PFs in 37 independent coordinates of p(i|o); all test points lie in the same affine span). Output `facets/pf.out.gz`.

The 3-party binary PF hull has exactly 29,344 facets, in 93 orbits under the 384-element symmetry group (`facets/classify.py`, `classes_full.json`). Orbit sizes: 384 (66 orbits), 192 (17), 128 (2), 96 (5), 64 (3). The 8 LP classes of D.7 are all among them: LP classes 0, 1, 2 are full orbits 82, 83, 84 (418 tight PFs); classes 3-6 are orbits 42, 38, 37, 41 (238); class 7 is orbit 91 (436).

Violations. E_{ex1} violates 24 of the 93 facet orbits, orbit 7 violates 53, orbit 8 violates 41, and every test point violates at least one. **Facet orbit 82, the class-0 inequality of D.7 p(000|000) ≤ p(000|001)+p(010|001)+p(000|100)+p(010|110)+p(011|111), is violated by E_{ex1} and by all 51 sampled non-deterministic vertex orbits** (by some member of the facet's orbit). So this single inequality class separates every sampled non-deterministic vertex from the PF hull. Proving it from purifiability is the natural next target. Status: facet list exact (normaliz); the derivation from purifiability is open.

D.8 Toward deriving facet orbit 82 from purifiability (Oct 10 2026): not proved

Sanity checks. Lugano and all PF mixtures satisfy every PF-hull facet, by construction. E_{ex1} violates orbit 82 (D.7).

Proposition D.8 [proved, negative]: no certificate built from the fixed source state alone can work. The proposal was an SDP over Gram matrices of sink vectors, with the source in |psi⟩ and parties applying arbitrary local unitaries (with ancillas, e.g. Hadamard superpositions of deterministic interventions). All such Gram data is a linear function of W. Inner products between outputs for ancilla inputs a, a' are $\mathrm{Tr}[W (M_a \otimes M_{a'}\dagger \ldots)]$. Unitarity on the psi-sector (orthonormal outputs for orthonormal ancilla inputs) follows from validity of W under the local operations that prepare the ancilla in a, a', (a+a')/sqrt2 and (a+ia')/sqrt2. By D.5 every classical process, including E_{ex1}, is valid under all such operations. So these constraints are satisfiable by E_{ex1}, and any SDP over psi-sector Gram data is feasible for it: no dual certificate exists. (This strengthens D.2 from classical to arbitrary local interventions.) Any derivation of the facet must use V on source states orthogonal to psi. In AFNB Thm 4 terms, that is the non-convex isotropic-subspace condition, the same obstruction met in the certificate work (App. C). **The facet has not been derived; no certificate was found.**

Numerical boundary test (`physics/boundary.py`, `boundary.json`). On $x_t = (1-t)$ Lugano + $t\, E_{ex1}$, the PF hull is left at $t_h \sim 2e\text{-}7$: Lugano is a vertex and this direction exits the hull at once. The full purification search (gpusearch, float64, 8 restarts) finds no purification at t = 0.02 (best residual 52.2; rank 19, dim V_W 88) or at t = 0.2 (60.1); the same code reaches about 1e-13 on Lugano itself. This is consistent with the conjecture, since the

purifiable set appears to end where the PF hull does along this line. It is numerical evidence, not a proof. Generalisation to other facet orbits: not attempted, since no certificate exists to generalise.

Appendix E. Symmetry-reduced moment relaxation (negative result)

Stabilisers in the 384-element group of party permutations and input/output relabelings: E_{ex1} and orbit 50: 48; orbit 7: 16; orbits 8 and 20: 8; Lugano: 6; orbits 10 and 46: 4; 11 orbits: 2; 34 orbits: 1. Symmetry helps most where certificates already exist, and not at all for most open orbits.

Relaxation. Variables P (m x m) and Y (m^2 x m^2), Hermitian, with Y standing for P ⊗ P and F the swap. A rank-r projector onto a totally isotropic subspace satisfies $Y \geq 0$, $0 \leq P \leq I$, tr P = r, Tr_2 Y = rP, $Tr_2(YF) = P$ (from $P^2 = P$), $P \otimes I - Y \geq 0$ and $Tr_1[(Q_t \otimes 1)YF] = P\, Q_t\, P = 0$ for every form. So infeasibility would certify non-purifiability, for complex subspaces. The constraints are covariant under the stabiliser, so symmetry reduction would be legitimate.

Result: too weak. For E_{ex1} (m = 27, r = 16) the unreduced problem is small enough to solve directly by cyclic projections (code/stochastic/rep/pocs.py). The solver was checked on controls: it reaches feasibility at r = 1, 2, 4, and a 4-dim isotropic subspace exists numerically. At r = 16 the relaxation is feasible with strict margins: min eig Y = 0.13, eig P in [0.58, 0.63], min eig of $P \otimes I - Y$ = 0.019. P is close to (16/27)·I. E_{ex1} is non-purifiable and certified by single-form inertia, so this relaxation is strictly weaker than the inertia test. Symmetry-reducing it is pointless. Stronger convex certificates would need higher moment levels or must encode rank.

Appendix F. Infinite and continuous alphabets (draft)

Literature (verified). Giacomini, Castro-Ruiz and Brukner extend the process-matrix framework to continuous-variable systems. Naive infinite-dimensional process matrices are singular, and they use a phase-space (Wigner) representation instead (New J. Phys. 18, 113026 (2016); https://arxiv.org/abs/1510.06345). The Baumeler-Wolf polytope work is for finite alphabets (https://arxiv.org/abs/1507.01714). The fixed-point characterisation of logical consistency (Baumeler and Wolf, https://arxiv.org/abs/1511.05444) and the 2025 flow-of-causal-structures paper (Phys. Rev. Research, doi:10.1103/jsmm-tq72) work with discrete alphabets. We found no treatment of process functions on infinite alphabets, or of their unitary lifts.

Setting. Finitely many parties. Each input alphabet I_k is a countable set carrying a group structure (e.g. Z), and the outputs O_k are countable. Hilbert spaces are l^2 of these sets. A process function has a unique fixed point for every choice of local functions, exactly as before.

F.1 [proved] The lift is unitary. $U|o,e\rangle = |w(o)+e, o\rangle$ is a bijection of basis states, with inverse $(i,f) \to (f, i - w(f))$. So it is a unitary on l^2, in any cardinality.

F.2 [proved] Matrix elements and rows. Let $V_k : I_k \otimes X_k \to O_k \otimes Y_k$ be unitaries, with X_k, Y_k countable ancilla bases (they may be shared or entangled across parties; unitarity is a property of the operator, so product bases suffice). Closing the loop with U gives matrix elements whose defining sum over (o, i) has exactly one nonzero term. So no convergence question arises: $\langle f,y|M|e,x\rangle = \langle f,y|\, V\, |w(f)+e, x\rangle$, with $V = \otimes_k V_k$. Define $u_{f,y} := \sum_{e,x} \mathrm{conj}(\langle f,y|M|e,x\rangle)\, |e,x\rangle = T_{-w(f)}\, V\dagger\, |f,y\rangle$, where T_a translates the source register by a. Each $u_{f,y}$ is a unit vector. For f ≠ f', the exclusivity lemma gives a party k with $f_k \neq f'_k$ and $w_k(f) = w_k(f')$, so $\langle u_{f,y}|u_{f',y'}\rangle$ has the factor $\langle f_k,y_k|f'_k,y'_k\rangle = 0$. For f = f' and y ≠ y' the factor is $\langle y|y'\rangle = 0$. Hence $\{u_{f,y}\}$ is an orthonormal set, and M†, defined

on the basis by $|f,y\rangle \to u_{f,y}$, extends to an isometry. So M is a bounded operator with $M\,M^\dagger = 1$. Only basis vectors and the standard extension of an isometry from an orthonormal basis are used; no strong limits are needed.

The exclusivity lemma holds for any cardinality. If $s \neq s'$ and no such k existed, then $w_k(s) \neq w_k(s')$ whenever $s_k \neq s'_k$. Choose local functions with $f_k(w_k(s)) = s_k$ and $f_k(w_k(s')) = s'_k$ (and f_k constant if $s_k = s'_k$). Both s and s' are then fixed points, a contradiction. Only two values of each f_k are prescribed.

F.3 [proved] M is unitary (the column side, F.4 in the earlier draft). An orthonormal set $\{u_{f,y}\}$ is complete iff $\sum_{f,y} |\langle e,x|u_{f,y}\rangle|^2 = 1$ for every basis vector $|e,x\rangle$ (Parseval). Now $\sum_{f,y} |\langle f,y|V|w(f)+e,x\rangle|^2 = \sum_f \prod_k q_k(f_k \mid w_k(f))$, where $q_k(o|i) := \sum_{y_k} |\langle o,y_k|V_k|i+e_k, x_k\rangle|^2$. For each i, $q_k(.|i)$ is a probability distribution on O_k, because V_k is unitary (its columns are unit vectors). Draw the local functions $f_k(i)$ independently, with law $q_k(.|i)$, for every k and i: a countable product of probability spaces. For a fixed o, the event $\{f_k(w_k(o)) = o_k \text{ for all } k\}$ has probability $\prod_k q_k(o_k|w_k(o))$. By Tonelli, $\sum_o \prod_k q_k(o_k|w_k(o)) = E[$ number of fixed points of $o \to f(w(o))\ ] = 1$, since every deterministic choice of local functions has exactly one fixed point. Hence each column has norm 1, the set is complete, and $M^\dagger M = 1$. **Every process function on countable alphabets (inputs carrying a group structure) lifts to a unitary process: U is unitary and so is every induced map.** This argument also gives a dimension-free proof in the finite case. *Machine-checked in Lean for finite alphabets:* `PFUL.Column.sum_prod_eq_one` (the fixed-point averaging identity, over any commutative ring), `PFUL.induced_colnorm` (unit columns), and `PFUL.induced_conjTranspose_mul_of_colnorm` ($M^H M = 1$ from orthonormal rows plus unit columns, with no dimension count). Standard axioms only. For countable alphabets, the column identity and unit column norms are machine-checked too: `PFUL.Countable.tsum_prod_eq_one` ($\mathbb{R}\geq 0\infty$ sums, via the product measure `Measure.infinitePi`; inputs of any cardinality) and `PFUL.Countable.induced_colnorm`. Row orthonormality (`PFUL.Countable.induced_rows`, using the factorisation lemma `PFUL.Countable.hasSum_pi_prod`) and the assembly are machine-checked as well: `PFUL.Countable.induced_unitary_l2` gives a linear isometric equivalence between the ℓ^2 spaces whose matrix is M. So in Lean M is unitary for countable alphabets. Exclusivity gives orthogonal rows, and logical consistency under stochastic interventions gives unit columns.

The earlier draft's special cases (permutation interventions; causally ordered w) are now corollaries. The earlier remark that two-party process functions are causal still holds for any cardinality, by the two-fixed-point argument: first w_k is independent of o_k, then w_A and w_B cannot both be non-constant. That remark is no longer needed. No counterexample search is needed either, because F.3 rules out proper co-isometries.

F.5 Continuous alphabets [partly definitional]. Take I_k a locally compact abelian group with Haar measure (R, U(1)), and w measurable. Then U is a measure-preserving bijection of O x P, so it is unitary on L^2 (F.1 carries over). The induced map, however, involves pointwise evaluation $\langle f|V\,T_{w(f)}$. That is undefined for general unitaries on L^2, so "unitary under all local unitaries" first needs a definition, for instance via the phase-space approach of Giacomini et al. For measure-preserving local bijections the induced map is the bijection $e \to f(e)$. It is measure-preserving in the causal case (Fubini), but this is open in the non-causal case.

Status. Proved (paper and Lean, finite and countable), for countable alphabets with a group structure on the inputs and finitely many parties: the lift is unitary under all local unitaries with ancillas (F.1-F.3). Open: continuous alphabets, where the induced map first needs a definition (F.5). The proof of F.3 suggests the natural definition, via

the kernel ⟨f|V|w(f)+e⟩ with Haar measure, and the same probabilistic argument would then need the measurable-selection version of "exactly one fixed point".

References

1. M. Dummett, Can an effect precede its cause?, Aristotelian Society Supplementary Volume 28, 27 (1954); Bringing about the past, Philosophical Review 73, 338 (1964).
2. M. Black, Why cannot an effect precede its cause?, Analysis 16, 49 (1956).
3. D. Lewis, The paradoxes of time travel, American Philosophical Quarterly 13, 145 (1976).
4. K. Gödel, An example of a new type of cosmological solutions of Einstein's field equations of gravitation, Reviews of Modern Physics 21, 447 (1949).
5. M. S. Morris, K. S. Thorne and U. Yurtsever, Wormholes, time machines, and the weak energy condition, Physical Review Letters 61, 1446 (1988).
6. J. Friedman, M. S. Morris, I. D. Novikov, F. Echeverria, G. Klinkhammer, K. S. Thorne and U. Yurtsever, Cauchy problem in spacetimes with closed timelike curves, Physical Review D 42, 1915 (1990).
7. F. Echeverria, G. Klinkhammer and K. S. Thorne, Billiard balls in wormhole spacetimes with closed timelike curves: classical theory, Physical Review D 44, 1077 (1991).
8. A. Carlini, V. P. Frolov, M. B. Mensky, I. D. Novikov and H. H. Solodukhin, Time machines: the Principle of Self-Consistency as a consequence of the Principle of Minimal Action, International Journal of Modern Physics D 4, 557 (1995).
9. J. Earman, Bangs, Crunches, Whimpers, and Shrieks: Singularities and Acausalities in Relativistic Spacetimes, Oxford University Press (1995).
10. D. Deutsch, Quantum mechanics near closed timelike lines, Physical Review D 44, 3197 (1991).
11. S. Aaronson and J. Watrous, Closed timelike curves make quantum and classical computing equivalent, Proceedings of the Royal Society A 465, 631 (2009).
12. C. H. Bennett, D. Leung, G. Smith and J. A. Smolin, Can closed timelike curves or nonlinear quantum mechanics improve quantum state discrimination or help solve hard problems?, Physical Review Letters 103, 170502 (2009).
13. S. Lloyd et al., Closed timelike curves via postselection: theory and experimental test of consistency, Physical Review Letters 106, 040403 (2011).
14. O. Oreshkov, F. Costa and Č. Brukner, Quantum correlations with no causal order, Nature Communications 3, 1092 (2012).
15. Ä. Baumeler and S. Wolf, The space of logically consistent classical processes without causal order, New Journal of Physics 18, 013036 (2016).
16. Ä. Baumeler, F. Costa, T. C. Ralph, S. Wolf and M. Zych, Reversible time travel with freedom of choice, Classical and Quantum Gravity 36, 224002 (2019), arXiv:1703.00779.
17. M. Araújo, A. Feix, M. Navascués and Č. Brukner, A purification postulate for quantum mechanics with indefinite causal order, Quantum 1, 10 (2017).
18. A. Vanrietvelde, N. Ormrod, H. Kristjánsson and J. Barrett, Consistent circuits for indefinite causal order, arXiv:2206.10042 (2022).
19. H. Price, Does time-symmetry imply retrocausality? How the quantum world says yes, Studies in History and Philosophy of Modern Physics 43, 75 (2012).
20. M. S. Leifer and M. F. Pusey, Is a time symmetric interpretation of quantum theory possible without retrocausality?, Proceedings of the Royal Society A 473, 20160607 (2017).
21. T. Maudlin, A tale of two tensors, or, how not to argue for retrocausality, preprint (2017).
22. E. Specker, Die Logik nicht gleichzeitig entscheidbarer Aussagen, Dialectica 14, 239 (1960).

23. H. Dourdent, A. Leitherer, E.-C. Boghiu, K. Simonov, R. Kunjwal and A. Acín, What makes a causal loop consistent?, arXiv:2609.39735 (2026).
24. N. Daher Ahmed and R. Kunjwal, Characterizing unitaries via quasi-process functions, arXiv:2610.00579 (2026).
25. J. Faye, Backward causation, Stanford Encyclopedia of Philosophy, substantive revision 28 October 2025.
26. L. M. Tolksdorf and R. Verch, Quantum physics, fields and closed timelike curves, Communications in Mathematical Physics 357, 319 (2018).
27. Ä. Baumeler, A. S. Gilani and J. Rashid, Unlimited non-causal correlations and their relation to non-locality, Quantum 6, 673 (2022).
28. G. Tobar and F. Costa, Reversible dynamics with closed time-like curves and freedom of choice, Classical and Quantum Gravity 37, 205011 (2020), arXiv:2001.02511.
29. H. Dourdent, K. Simonov, A. Leitherer, E.-C. Boghiu, R. Kunjwal, S. Halder, R. Augusiak and A. Acín, Paradox-free classical non-causality and unambiguous non-locality without entanglement are equivalent, arXiv:2512.23599 (2025).
30. Ä. Baumeler, Causal loops: logically consistent correlations, time travel, and computation, PhD thesis, Università della Svizzera italiana (2017).